

Deleting duplicate elements from array:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 17 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h> #include<conio.h>
void main()
{
int a[100],n,i,j,k; clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++)
{
for(j=i+1;j<n;j++)
{
if(a[i]==a[j]){for(n--, k=j;k<n;k++)a[k]=a[k+1];j--;}
}
}
printf("Elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

Enter array size 1-100 9
Enter 9 elements 1 2 3 1 2 3 1 2 3
Elements are 1 2 3_

```

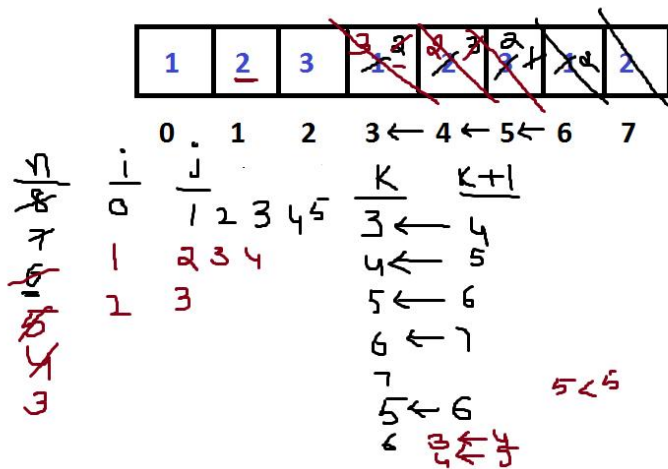
Enter array size 1-100 5
Enter 5 elements 1 1 1 1 1
Elements are 1_

```

```

for( i=0; i<n;i++ )
{
for( j=i+1; j<n;j++ )
{
if( a[i]==a[j] )
{
for(n-- ,k=j ;k<n; k++ ) a[k]=a[k+1];
}
}
}

```



2nd max, 2nd min array elements:

```
TC
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Line 2 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,j,t; clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<=n-2;i++)
{
for(j=i+1;j<=n-1;j++)
{
if(a[i]>a[j]){t=a[i];a[i]=a[j];a[j]=t;}}
printf("Elements ");for(i=0;i<n;i++)printf("%4d",a[i]);
for(i=1;i<n;i++)if(a[i]>a[0]){printf("\n2nd min=%d\n",a[i]);break;}
for(i=n-2;i>=0;i--)if(a[i]<a[n-1]){printf("2nd max=%d",a[i]);break;}
getch();
}
}
```

Enter array size 1-100 9
Enter 9 elements 3 0 1 6 4 8 6 2 -1
Elements -1 0 1 2 3 4 6 6 8
2nd min=0
2nd max=6_

```
TC
Enter array size 1-100 9
Enter 9 elements 1 1 1 2 3 4 5 5 5
Elements      1   1   1   2   3   4   5   5   5
2nd min=2
2nd max=4
```

NTH MIN, NTH MAX ARRAY ELEMENTS:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 1 Col 19 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h> #include<conio.h>
void main()
{
int a[100],n,i,j,t,max, min; clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter nth min, nth max values "); scanf("%d%d",&min,&max);
for(i=0;i<=n-2;i++)
{
for(j=i+1;j<=n-1;j++)
{
if(a[i]>a[j]){t=a[i];a[i]=a[j];a[j]=t;}}
printf("Elements ");for(i=0;i<n;i++)printf("%4d",a[i]);
for(i=1;i<n;i++)
{if(a[i]>a[i-1])min--;
if(min==1){printf("\nmin=%d\n",a[i]);break;}}
for(i=n-2;i>=0;i--)
{if(a[i]<a[i+1])max--;if(max==1){printf("max=%d",a[i]);break;}}
getch();
}
}
```

Enter array size 1-100 9
Enter 9 elements 1 2 3 4 5 6 7 7 8
Enter nth min, nth max values 3 4
Elements 1 2 3 4 5 6 7 7 8
min=3
max=5

```
TC
Enter array size 1-100 9
Enter 9 elements 3 0 -2 7 -5 9 1 4 6
Enter nth min, nth max values 4 5
Elements   -5  -2   0   1   3   4   6   7   9
min=1
max=3
```

```
TC
Enter array size 1-100 9
Enter 9 elements 1 1 2 2 4 5 6 7 7
Enter nth min, nth max values 3 3
Elements    1   1   2   2   4   5   6   7   7
min=4
max=5
```

```

for(i=1; i<n; i++)
{
  if(a[i]>a[i-1]) min--;
  if(min==1) p(min=a[i]); break;
}
for(i=n-2; i>=0; i--)
{
  if(a[i]<a[i+1]) max--;
  if(max==1) p(max=a[i]); break;
}

```

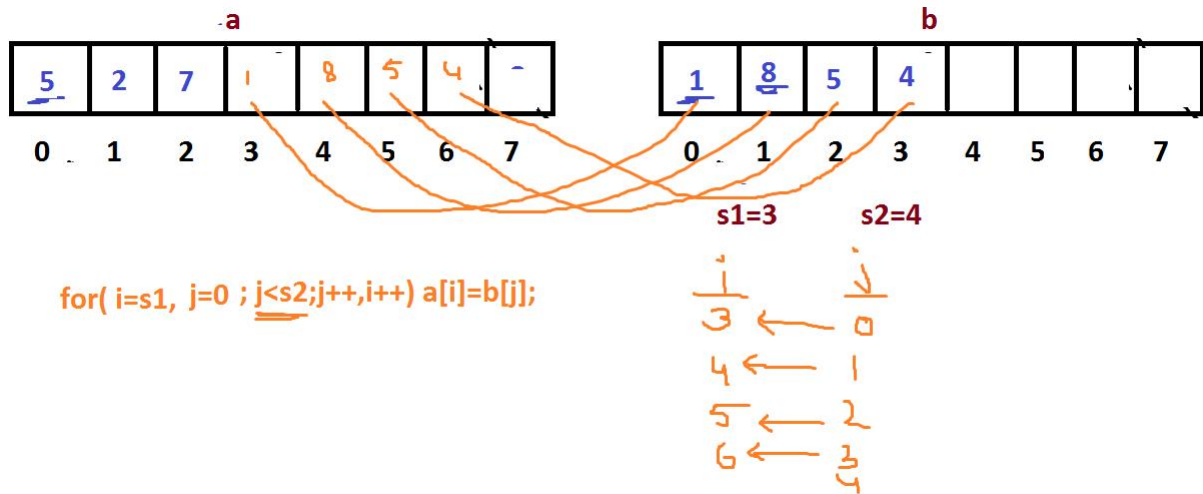
<u>1</u>	2	3	4	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>
0	1	2	3	4	5	6	7

<u>i</u>	<u>max</u>	<u>min</u>		<u>i</u>
1	1	2	3RD MIN = 3 4 MAX = 5	6
2	2	2		5
	1	1		4 ✓

Merging of array:


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 2 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],b[100],t,s1,s2,i,j; clrscr();
printf("Enter a, b arrays size 1-100 ");scanf("%d %d",&s1, &s2);
printf("Enter %d elements for a array ",s1);
for(i=0;i<s1;i++)scanf("%d",&a[i]);
printf("Enter %d elements for b array ",s2);
for(i=0;i<s2;i++)scanf("%d",&b[i]);
for( i=s1,j=0; j<s2; i++, j++) a[i]=b[j];
for(i=0;i<s1+s2;i++)
{
for(j=i+1;j<s1+s2;j++)
{
if(a[i]>a[j]){t=a[i];a[i]=a[j];a[j]=t;}}
printf("Elements ");for(i=0;i<s1+s2;i++)printf("%4d",a[i]);
getch();
}
```

Enter a, b arrays size 1-100
3 5
Enter 3 elements for a array 8 -7 4
Enter 5 elements for b array 1 0 4 -3 9
Elements -7 -3 0 1 4 4 8 9



Frequency of array elements:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 3 Col 19 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h> #include<conio.h>
void main()
{
int a[100],b[100]={0},n,i,j,c; clrscr();
printf("Enter arrays size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<n;i++)
{
if(b[i]!=-1)
{
for(c=1, j=i+1;j<n;j++)
{
if(a[i]==a[j]){c++; b[j]=-1;}
}
b[i]=c;
}
}
for(i=0;i<n;i++)if(b[i]!=-1)printf("%d found %d times\n",a[i],b[i]);
getch();
}
```

Enter arrays size 1-100 9
Enter 9 elements 1 2 9 8 1 4 2 1 3
1 found 3 times
2 found 2 times
9 found 1 times
8 found 1 times
4 found 1 times
3 found 1 times

```

Enter arrays size 1-100 10
Enter 10 elements 9 4 4 0 1 2 3 1 2 3
9 found 1 times
4 found 2 times
0 found 1 times
1 found 2 times
2 found 2 times
3 found 2 times

```

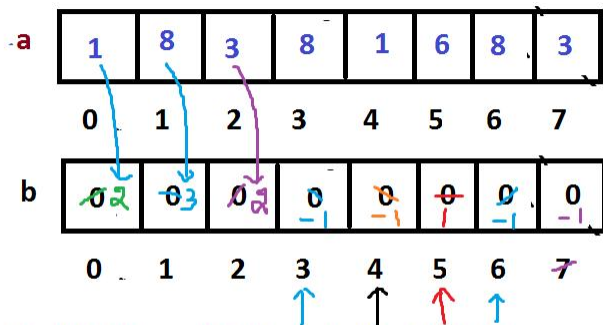
```

for(i=0; i<8; i++)
{
    if( b[i] != -1 ) {
        for(c=1, j=i+1; j<9; j++)
        {
            if(a[i] == a[j]) { c++, b[j] = -1; }
        }
        b[i] = c;
    }
}

```

Handwritten annotations for the code above:

- For the first loop, i values 0 to 7 are written above the loop body, and j values 1 to 8 are written above the inner loop body.
- For the inner loop, c values 1 to 8 are written above the loop body.
- For the inner loop, j values 1 to 8 are written above the loop body.
- For the inner loop, $a[i]$ and $a[j]$ are written above the comparison.
- For the inner loop, $c++$ and $b[j] = -1$ are written above the assignment.
- For the inner loop, $b[i] = c$ is written above the assignment.



for(i=0; i<8; i++) if(b[i] != -1) p("%d found %d times\n", a[i], b[i]);

Two dimensional arrays:

Array with several rows and columns.

Array with two subscripting operators **[] []**.

It is array of arrays. i.e. collection of one-dimensional arrays.

It is implicit double pointer.

It is a $n \times n$ matrix.

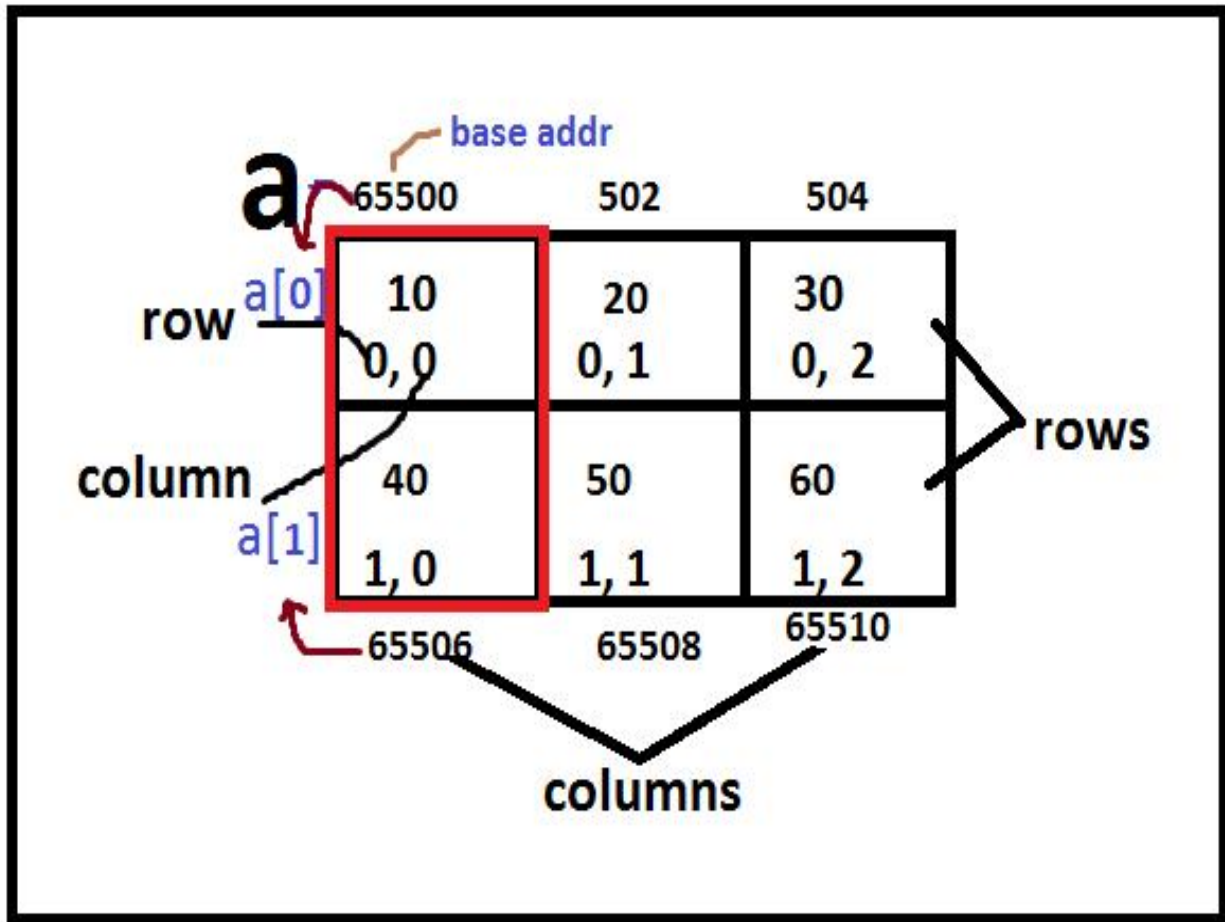
Syntax:

datatype variable [rows] [columns] = {elements} ;

Eg:

```
int a[ 2 ][ 3 ] = { { 10, 20 , 30 } , { 40 ,50, 60 } };
```

stack



In two dimensional array the rows/first subscript is working as array of pointers and they stores first column address of each row. Hence it is an implicit/internal double pointer.

In the above example, To print the first row, first column value, we have to use

`printf("%d", a[0][0]);` → 10

Internally how this statement is working ?

`a[0]` means value at `a[0]` i.e. 65500.

`65500 + [0] col` → `65500 + 0 * 2` → `65500` →
value at 65500 is 10.

Index no

Int size

stack

<code>a[1][2]</code>	<u>60</u>	65500
<code>a[1][1]</code>	<u>50</u>	65498
<code>a[1][0]</code>	40	65496
<code>a[0][2]</code>	<u>30</u>	65494
<code>a[0][1]</code>	<u>20</u>	65492
<code>a[0][0]</code>	<u>10</u>	65490

`int a[2][2]={10,20,30,40};`

<code>a[1][1]</code>	40	65496
<code>a[1][0]</code>	<u>30</u>	65494
<code>a[0][1]</code>	<u>20</u>	65492
<code>a[0][0]</code>	<u>10</u>	65490

```
int a[2][3]={{1,2,3},{4,5,6}};
```

